

ENTERPRISE PRODUCT DOCUMENTATION

Daily Tracker v2.0

Comprehensive End-to-End Feature & Architecture Manual

An enterprise-grade attendance, task orchestration, AI copilot, and human resource management platform built with .NET 10 Web API and React 18.

System Specifications & Production State

- Architecture: Distributed REST API + React SPA + Real-time WebSockets
- Backend Engine: ASP.NET Core (.NET 10), EF Core 10, SQL Server, Redis
- Frontend Client: React 18, TypeScript, Tailwind CSS, TanStack Query, Vite
- AI Engine: Google Gemini 2.0 Flash with Context-Aware Agent Automation
- Release Status: Production Ready (Main Branch Audited - 0 Errors)

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01. Platform Architecture & Security

Enterprise Full-Stack Infrastructure Overview

System Design & Component Tiering

Daily Tracker v2.0 is engineered as a high-throughput, enterprise-ready workforce operations management system. The platform bridges real-time employee activity tracking with modern HR workflows and intelligent automation powered by generative artificial intelligence.

- **API Gateway & Services:** Built on modern .NET 10 Web API utilizing Entity Framework Core 10 for ORM, SQL Server for relational persistence, and Redis for distributed caching and session state management.
- **Client Application:** Engineered with React 18, TypeScript, Tailwind CSS, and Vite. Leverages TanStack React Query for background cache revalidation, optimistic updates, and offline service worker caching via PWA.
- **Real-Time Event Bus:** Powered by ASP.NET Core SignalR hubs mounted at /hubs/chat and /hubs/notifications, streaming live presence updates, unread counts, and instant messages with zero polling overhead.
- **Security & Session Boundaries:** Strict authentication architecture using JSON Web Tokens (JWT) stored in Secure, HttpOnly, SameSite cookies to completely eliminate Cross-Site Scripting (XSS) token exfiltration risks.

Role-Based Access Control (RBAC)

The authorization matrix enforces strict boundaries across three core organizational roles:

- **Employee:** Access to personal dashboard, clock-in/out, task logging, goal tracking, leave/WFH applications, peer kudos, chat, and AI Copilot.
- **Manager:** Complete visibility over team attendance, leave/WFH approval dashboard, EOD daily reviews, 1-click email action callbacks, and employee audit analytics.
- **Administrator:** Full tenant control, role assignment, system announcements, department configuration, facial recognition registration approval, and payroll salary adjustments.

02. Authentication & Biometrics

Zero-Trust Identity, TOTP 2FA & Face Verification

Multi-Factor Authentication (MFA / 2FA)

The security layer guarantees multi-layered defense against credential stuffing and unauthorized terminal access:

- **Time-Based One-Time Password (TOTP):** Employees can link standard authenticator apps (Google Authenticator, Microsoft Authenticator) via dynamic QR code generation powered by QRCodeR.
- **Email OTP Verification:** Transactional email codes for registration activation, password recovery, and unfamiliar browser logins.
- **Trusted Device Recognition:** Secure browser fingerprinting allowing users to trust authorized workstations for 30 days while enforcing MFA on foreign devices.

Biometric Face Recognition Engine

To prevent proxy attendance ("buddy punching"), the platform incorporates client-side biometric facial landmark verification:

- **Neural Vector Embeddings:** Captures 128-dimensional facial vector descriptors during one-time enrollment in the secure Face Setup module.
- **Liveness & Verification:** Matches the live camera feed during check-in against registered descriptors with euclidean distance threshold matching (< 0.45).
- **Security Audit Logging:** Every biometric verification attempt—both successful and failed—is logged with timestamp, confidence score, and IP address for managerial inspection.

03. Attendance, Geofencing & Breaks

Precision Time Tracking & Location Compliance

Geofenced Office Compliance

The check-in/check-out pipeline enforces GPS boundary compliance to ensure physical presence at designated office facilities:

- **Haversine Distance Engine:** Calculates exact distance between employee geolocation and designated corporate campus coordinates.
- **Configurable Office Radius:** Enforces configurable radius tolerance (e.g., 50 meters). Check-in attempts outside the boundary return 403 Forbidden with real-time distance feedback.
- **Smart Exception Handling:** Automatically disables geofencing checks for employees on approved Work-From-Home (WFH), half-day schedules, recognized public holidays, or weekend shifts.

Break Management & Net Work Hours

Work duration is tracked with mathematical accuracy to guarantee transparent payroll and overtime records:

- **Multi-Category Breaks:** Employees can toggle Tea, Lunch, Coffee, or Personal breaks with real-time running timers.
- **Auto-Close on Checkout:** Checking out automatically finalizes any ongoing active break, computes duration, and recalculates total break minutes.
- **Net Work Calculation:** $\text{Total Work Minutes} = \text{Max}(0, \text{Elapsed Clock Time} - \text{Total Break Minutes})$. Accidental double check-out submissions are strictly guarded.

04. Task Management & Kanban Board

Agile Execution, Time Logging & Task Timers

Interactive Task Lifecycle

The task tracking module enables employees to log, categorize, and prioritize their daily deliverables seamlessly:

- **State Transitions:** Full support for four operational states: InProgress, Completed, Blocked, and OnHold.
- **Kanban & List Views:** Switch between high-density list views and a fluid visual Kanban board with drag-and-drop status transitions.
- **Granular Metadata:** Tracks project associations, estimated vs actual minutes spent, priority levels (High, Medium, Low), and searchable tags.

Automated Task Timers & Templates

- **Active Task Timer:** Live stopwatch component allowing employees to start/pause timers against specific tasks, automatically updating time spent.
- **Recurring Task Templates:** Create reusable task templates for routine daily or weekly tasks. 1-click execution clones templates directly into today's log.

05. Daily Goals & Productivity

Target Setting, Metrics & Trend Analytics

Daily Goal Setting & Real-Time Progress

Daily Goals align individual employee output with team sprint objectives:

- **Custom Targets:** Set target completed tasks (e.g., 5 tasks) and target active work hours (e.g., 8 hours) each morning.
- **Live Progress Engine:** Real-time visual progress ring and completion percentage dynamically updating as tasks are completed and hours logged.
- **Productivity Score Algorithm:** Proprietary scoring model factoring task completion ratio, hours logged vs target, and break adherence.

Goal History & Trend Analytics

The dedicated Goal History portal provides actionable retrospective data:

- **Multi-Period Aggregation:** Instant filters for 7-day, 14-day, and 30-day productivity trends.
- **Detailed Daily Breakdowns:** Inspect historical daily targets vs actual achievements, tasks completed, and productivity grades.

06. End of Day (EOD) Reporting

Structured Daily Summaries & Manager Reviews

Structured EOD Report Format

End of Day reports provide structured accountability without administrative overhead:

- **Accomplishments ("What was done"):** Comprehensive breakdown of deliverables achieved during the shift.
- **Blockers & Impediments:** Immediate flag for unresolved technical or operational roadblocks requiring managerial intervention.
- **Tomorrow's Plan:** Advance agenda and priorities lined up for the subsequent working day.
- **Learnings & Mood Rating:** Self-reported daily learnings and 5-point mood index (Great, Good, Okay, Tired, Stressed).

AI Auto-Generation Engine

Employees can generate their entire EOD draft in 1 click via the integrated Gemini 2.0 AI engine:

- **Automatic Synthesis:** The AI inspects today's checked-in time, calculated net work hours, completed tasks, and peer support logs.
- **Dual-Interface Availability:** Available on the dedicated EOD Report Page (/eod-reports) and directly inside the quick Team Dashboard modal.

Manager Review & Rating Portal

- **Team Submission Queue:** Managers can review pending reports, provide constructive feedback, assign performance ratings, and flag outstanding contributions.

07. Remote Work & Half-Day Workflows

Automated Routing & 1-Click Email Approvals

Flexible Remote Work Submission

Empowers hybrid and distributed teams with smooth remote work administration:

- **Granular Scheduling:** Support for Full-Day WFH, First-Half Half-Day (morning slot), and Second-Half Half-Day (afternoon slot).
- **Future Date Picker:** Plan remote days in advance with automatic validation preventing duplicate or past-date submissions.

1-Click Email Approval System

Removes managerial bottlenecks via cryptographic tokenized email actions:

- **Automated Manager Email:** Upon submission, the employee's direct manager receives an immediate HTML notification containing request details.
- **1-Click Decision Links:** Secure Approve and Reject buttons inside the email execute the status change instantly via tokenized endpoints without requiring VPN or login.
- **Calendar Integration:** Approved WFH and Half-Day entries automatically populate the Team Calendar and adjust geofencing rules.

08. Annual Leave Entitlement

Quotas, Working-Day Calculations & Balance Vault

Annual Entitlement Engine

Transitioned from legacy monthly caps to a comprehensive annual allocation structure:

- **Leave Classifications:** Casual Leave (12 days/yr), Sick Leave (7 days/yr), Earned Leave (15 days/yr), Compensatory Off (unlimited), and Unpaid Leave.
- **Real-Time Quota Verification:** System validates requested duration against remaining annual balance prior to application acceptance.

Working-Day Calculation Engine

Accurate date range calculation guarantees employees are never charged leave days unfairly:

- **Weekend Filtering:** Automatically excludes Saturdays and Sundays from multi-day date spans.
- **Public Holiday Exclusion:** Cross-references company holiday schedules from the database, deducting public holidays from requested leave periods.
- **Real-Time Balance Dashboard:** Transparent visual breakdown displaying Entitlement, Used, Pending Approval, and Remaining days per leave category.

09. Real-Time Chat & Social Recognition

Live SignalR Communication & Kudos Leaderboards

Enterprise SignalR Chat Hub

Fosters instant collaboration across distributed teams:

- **Direct & Group Conversations:** 1-on-1 private messaging and multi-member team group channels.
- **Rich Collaboration Features:** Typing indicators, message read receipts, emoji reactions, and secure file attachments.
- **Global Announcements:** HR and Admins can publish pinned priority announcements with read acknowledgment tracking.

Peer Kudos & Recognition Board

Boosts team morale through positive peer recognition:

- **Categorized Recognition:** Award kudos tagged with core values (#TeamPlayer, #Innovation, #ProblemSolver, #CustomerFirst).
- **Monthly Leaderboards:** Real-time gamified leaderboard highlighting the most recognized employees across the organization.

10. Meeting Logs & Action Items

Collaborative Agendas & Follow-up Execution

Meeting Lifecycle Management

Eliminates unproductive meetings by connecting schedules directly with accountability:

- **Scheduling & Invites:** Create meetings with rich agendas, scheduled start/end times, and attendee invitations.
- **RSVP Status Tracking:** Live tracking of attendee responses (Accepted, Declined, Tentative).

Action Items & Deliverables

Every meeting produces tangible follow-ups:

- **Assigned Deliverables:** Attach actionable items to specific meeting logs with assignees and explicit deadlines.
- **Completion Tracking:** Assignees can mark action items as InProgress or Completed directly from their dashboard.

11. Gemini 2.0 Flash AI Copilot

Context-Aware Agent with Voice Interaction

Live Organizational Context

Powered by Google's Gemini 2.0 Flash generative AI model, the Copilot possesses live, authorized access to user data:

- **Context-Aware Intelligence:** Instantly answers questions about today's work hours, logged deliverables, leave balances, and upcoming meetings.
- **Smart Local Fallback:** Includes a resilient rule-based inference engine that continues serving user requests even if external API limits are reached.

Conversational Action Execution

The Copilot does not just answer questions—it executes operational workflows via natural language:

- **Executable Actions:** "Start my tea break", "Clock me out", "Apply for WFH tomorrow", "Create a high priority task", or "Draft my EOD report".
- **Safety Action Cards:** Presents confirmation cards in the chat with clear parameters before triggering backend operations.
- **Voice Recognition:** Speech-to-text input microphone allowing seamless voice dictation with intelligent transcript appending.

12. HR, Payroll & Employee Lifecycle

Vault, Performance Reviews & Digital Offboarding

Document Vault & Compliance

- **Secure Storage:** Categorized repository for contracts, identification documents, policy manuals, and tax filings.
- **Access Controls:** Employees can view and download their verified records with role-restricted administrative uploads.

Performance Review Cycles

- **Multi-Stage Cycles:** Configurable review cycles featuring employee self-assessment, manager evaluations, KPI scores, and final sign-offs.

Payroll & Salary Slips

- **Detailed Compensation:** Transparent breakdown of Basic Pay, HRA, Allowances, PF, Tax Deductions, and Net Salary.
- **PDF Payslip Downloads:** Generate and download official PDF payslips directly from the portal.

Digital Resignation & Offboarding

- **Structured Exit Process:** Formal resignation submission with notice period calculation and manager review.
- **Multi-Department Checklist:** Interactive handover checklist covering IT hardware return, finance clearance, and knowledge transfer sign-offs.

